course	unique	days	time	room	instructor

Graduate standing, and Computer Science 381K or equivalentknowledge of artificial intelligence and LISP. Taught as Web-based course. meets with A I 391L.

•	52704				KLIVANS, A

A I 394D DEEP LEARNING-WB

Hour(s) to be arranged. Explore the basic building blocks and intuitions behind designing, training, tuning, and monitoring of deep networks. Examine both the theory of deep learning, as well as hands-on implementation sessions in pytorch. Explore a series of application areas of deep networks in: computer vision, sequence modeling in natural language processing, deep reinforcement learning, generative modeling, and adversarial learning. Only one of the following may be counted: Artificial Intelligence 394D, Computer Science 394D, Data Science 394D, 395T (Topic: Deep Learning). Prerequisite: Graduate standing, and experience in artificial intelligence and machine learning. Restricted to students in the Master of Science in Artificial Intelligence Online program. Taught as a Web-based course. meets with DSC 394D, C S 394D, A I 394D.

•	52705				KRAEHENBUEHL, P

A I 395T ADV IN DEEP GEN MODELS-WB

Hour(s) to be arranged. Course number may be repeated for credit when the topics vary. Prerequisite: Graduate standing. Taught as a Web-based course.

•	52708				LIU, Q

A I 394D DEEP LEARNING-WB

Hour(s) to be arranged. Explore the basic building blocks and intuitions behind designing, training, tuning, and monitoring of deep networks. Examine both the theory of deep learning, as well as hands-on implementation sessions in pytorch. Explore a series of application areas of deep networks in: computer vision, sequence modeling in natural language processing, deep reinforcement learning, generative modeling, and adversarial learning. Only one of the following may be counted: Artificial Intelligence 394D, Computer Science 394D, Data Science 394D, 395T (Topic: Deep Learning). Prerequisite: Graduate standing, and experience in artificial intelligence and machine learning. Taught as a Web-based course. meets with A I 394D.

•	52709				KRAEHENBUEHL, P

A I 395T ADVANCES IN DEEP LEARNING-WB

Hour(s) to be arranged. Course number may be repeated for credit when the topics vary. Prerequisite: Graduate standing. Restricted to students in the Master of Science in Artificial Intelligence Online program. Taught as a Web-based course. meets with DSC 395T, A I 395T.

•	52710				KRAEHENBUEHL, P

ADVANCES IN DEEP LEARNING-WB

Hour(s) to be arranged. Course number may be repeated for credit when the topics vary. Prerequisite: Graduate standing. Taught as a Web-based course. meets with A I 395T.

•	52714				KRAEHENBUEHL, P

THESIS A-WB

Hour(s) to be arranged. Course number may be repeated for credit when the topics vary. Prerequisite: Graduate standing. Restricted to students in the Master of Science in Artificial Intelligence Online program. Taught as a Web-based course.

•	52715				

THESIS B-WB

Hour(s) to be arranged. Course number may be repeated for credit when the topics vary. Prerequisite: Graduate standing. Restricted to students in the Master of Science in Artificial Intelligence Online program. Taught as a Web-based course.

•	52720				

C S 303E ELEMS OF COMPUTERS/PROGRAMMING

Designed to accommodate 100 or more students. Problem solving and fundamental algorithms for various applications in science, business, and on the World Wide Web, and introductory programming in a modern object-oriented programming language. Only one of the following may be counted: Computer Science 303E, 312, 312H. Credit for Computer Science 303E may not be earned after a studenthas received credit for Computer Science 314, or 314H. May not be counted toward a degree in computer science.

•	52775	MW	1100 - 1230P	WEL 1.308	RAVISHANKAR, V
•	52780	MW	1230 - 200P	WEL 1.308	RAVISHANKAR, V

ELEMS OF COMPUTERS/PROGRAMMING

Designed to accommodate 100 or more students. Additional hour(s) to be arranged. Problem solving and fundamental algorithms for various applications in science, business, and on the World Wide Web, and introductory programming in a modern object-oriented programming language. Only one of the following may be counted: Computer Science 303E, 312, 312H. Credit for Computer Science 303E may not be earned after a studenthas received credit for Computer Science 314, or 314H. May not be counted toward a degree in computer science. Partially taught as a Web-based course.

•	52785	F	900 - 1000	WEL 1.308	YOUNG, W
•	52790	F	1000 - 1100	WEL 1.308	YOUNG, W
•	52795	F	1100 - 1200	WEL 1.308	YOUNG, W

C S 103F ETHICAL FNDTN COMP SCIENCE

Designed to accommodate 100 or more students. An introduction to reasoning about the effects of the technology createdby computer scientists and understanding the ethical and real-world consequences of that technology. Emphasizes direct participation and discussion using case studies to illustrate ethical topics. Computer Science 103F and 109 (Topic: Ethical Foundations of Computer Science) may not both be counted.

•	52800	T	100 - 200P	GAR 0.102	QUIMBY, M

course	unique	days	time	room	instructor

C S 104C COMPETITIVE PROGRAMMING

Explores the algorithms needed to solve a given problem, reason about corner cases, and write correct implementations under time pressure. Focuses on skills useful for international programming competitions, whiteboard job interviews, and programming tasks that require quickly solving challenging problems with creative use of algorithms. Computer Science 104 and 109 (Topic: Competitive Programming) may not both be counted. Prerequisite: Computer Science 311 or 311H with a grade of at least C-, and registration or credit in Computer Science 314 or 314H with a grade of at least C-.

•	52805	M	400 - 500P	RLP 0.128	VOUGA, P

C S 309 AUTONOM INTELLGNT ROBTCS I-FRI

Designed to accommodate 35 or fewer students. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to students in the Freshman Research Initiative.

•	52810	TTH	330 - 500P	WEL 2.306	HART, J

AUTONOMOUS DRIVING-FRI

Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic.

•	52815	TTH	330 - 500P	GDC 1.406	DEPUE WARD, L

BIG DATA IN BIOLOGY-FRI

Restricted enrollment; contact the department for permission to register for this class. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to students in the Freshman Research Initiative. meets with INB 321G.

•	52820	TTH	1100 - 1230P	FNT 1.104	ARASAPPAN, D

GEOMETRY OF SPACE-FRI

Designed to accommodate 35 or fewer students. Restricted enrollment; contact the department for permission to register for this class. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to students in the Freshmen Research Initiative.

•	52825	TTH	330 - 500P	WEL 3.320	MITRA, S

MEDICAL ROBOTICS-FRI

Designed to accommodate 35 or fewer students. Restricted enrollment; contact the department for permission to register for this class. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to the students in the Freshman Research Initiative.

•	52830	MW	330 - 500P	BUR 208	DEPUE WARD, L

ROBOT LEARNING-FRI

Designed to accommodate 35 or fewer students. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to students in the Freshmen Research Initiative.

•	52835	TTH	330 - 500P	WEL 3.310	XU, J

SPACE/ROBOTICS-FRI

Designed to accommodate 35 or fewer students. Additional hour(s) to be arranged. Restricted enrollment; contact the department for permission to register for this class. Course number may be repeated for credit when the topics vary. Prerequisite: Varies with the topic. Restricted to the students in the Freshman Research Initiative.

•	52840	MW	330 - 500P	PMA 5.114	DEPUE WARD, L

1-QUANTUM COMPUTING I-FRI

Course number may be repeated for credit when the topics vary. Introduction to quantum information and communication. Computer Science 309 (Topic: Quantum Computing-FRI) and 309 (Topic 1) may not both be counted. Prerequisite: Varies with the topic. Restricted to students in the Freshman Research Initiative. meets with PHY 208F.

•	52845	TTH	330 - 500P	CBA 4.344	LA COUR, B

C S 312 INTRODUCTION TO PROGRAMMING

Designed to accommodate 100 or more students. First part of a two-part sequence in programming. Fundamental concepts of structured programming; procedures and data structures with a focus on problem solving strategies and implementation; introduction to concepts of informal specification, informal reasoning about program behavior, debugging, and ad hoc testing. Only one of the following courses may be counted: Computer Science 303E,312, 312H. Credit for Computer Science 312 may not be earned after a student has received credit for Computer Science 314 or 314H. Prerequisite: Credit with a grade of at least C- or registration for Mathematics 408C, 408K, or 408N.

•	52905	MWF	900 - 1000	GDC 2.216	RAMSEY, C
		M	1100 - 1200	SZB 3.814	
•	52910	MWF	900 - 1000	GDC 2.216	RAMSEY, C
		M	200 - 300P	RLP 0.112	
•	52915	MWF	900 - 1000	GDC 2.216	RAMSEY, C
		M	300 - 400P	ART 1.110	
•	52920	MWF	900 - 1000	GDC 2.216	RAMSEY, C
		M	400 - 500P	RLP 1.106	
•	52925	MWF	1000 - 1100	GDC 2.216	RAMSEY, C
		M	1100 - 1200	WCP 5.102	
•	52926	MWF	1000 - 1100	GDC 2.216	RAMSEY, C
		M	1200 - 100P	GDC 1.406	
•	52927	MWF	1000 - 1100	GDC 2.216	RAMSEY, C
		M	100 - 200P	GAR 3.116	
•	52928	MWF	1000 - 1100	GDC 2.216	RAMSEY, C
		M	200 - 300P	ETC 2.102	

As of 10/03/2022